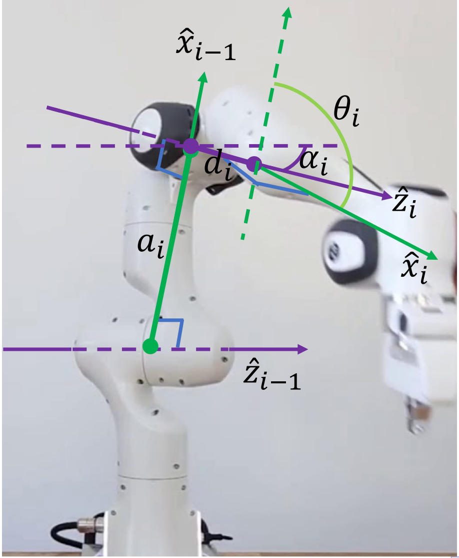
3D kinematics - learning outcomes

- Know how to use Denavit-Hartenberg convention to transform vectors from one coordinate frame to another.
- Know how to use symbolic Matlab codes to derive the compound homogeneous transformation matrix and the Jacobian matrix of a multi degree of freedom robot.
- Know how to apply the knowledge to both rigid link and robots with constant curvature continuum links.

3D kinematics



Frame assignment convention

Link frame attachment procedure.

1. Identify the joint axes and draw lines along them.
2. Identify the common perpendiculars between them, or points of intersection.
3. At the point of intersection, or at the point where the common perpendicular meets the ith axis, assign the link frame origin.
4. Assign the $\hat{z}_i$ axis pointing along the ith joint axis following the right hand rule (how a screw moves).
5. Assign $\hat{x}_i$ axis pointing along the common perpendicular, or if the $\hat{z}_i$ and $\hat{z}_{i+1}$ axes intersect, assign $\hat{x}_i$ to the normal to the plane containing the two axes.
6. Assign the $\hat{y}_i$ to complete the right-hand coordinate system (thumb points at $\hat{z}_i$, index finger points at $\hat{x}_i$, and the middle finger points at $\hat{y}_i$).

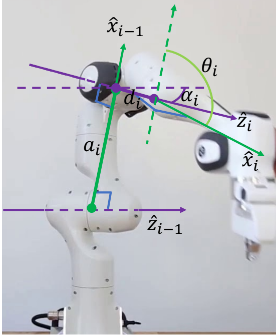
The base frame $\{0\}$ doesn't move.

1. Since frame $\{0\}$ is arbitrary, it is simple to choose $\hat{z}_0$ along the direction of axis-1.
2. Orient frame $\{0\}$ so that frame $\{1\}$ coincides with frame $\{0\}$ when the joint angle is zero.

The last frame $\{N\}$ is also arbitrary. Choose $\hat{x}_N$ freely, but so as to cause as many linkage parameters as possible to zero.

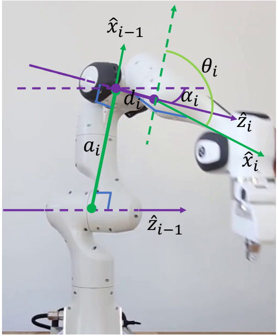
3D kinematics

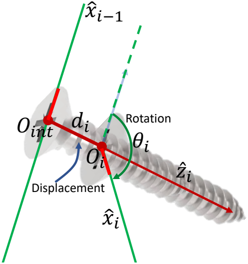
In this module, we follow the convention given in Khalil and Dombre.



1. a_i = the distance from $\hat{z}_{i-1}$ to $\hat{z}_i$ measured along $\hat{x}_{i-1}$.
2. α_i = the angle between $\hat{z}_{i-1}$ and $\hat{z}_i$ measured about $\hat{x}_{i-1}$.
3. d_i = the distance from $\hat{x}_{i-1}$ to $\hat{x}_i$ measured along $\hat{z}_i$.
4. θ_i = the angle between $\hat{x}_{i-1}$ to $\hat{x}_i$ measured about $\hat{z}_i$.

Please watch this video for an animation of DH parameter assignment.



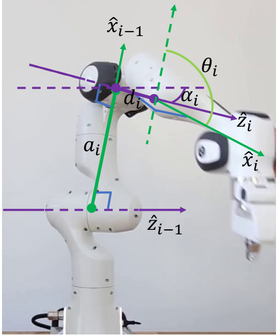


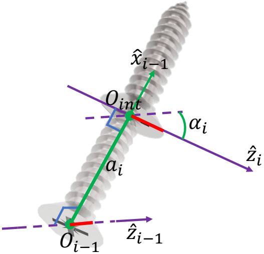
Then,

$$ {}^{i-1}\boldsymbol{T}_i = \boldsymbol{R}_{\hat{x}_{i-1}}(\alpha_i)\ \boldsymbol{D}_{\hat{x}_{i-1}}(a_i)\ \boldsymbol{R}_{\hat{z}_i}(\theta_i)\ \boldsymbol{D}_{\hat{z}_i}(d_i) \quad (1)$$

$$ {}^{i-1}\boldsymbol{T}_i = \mathrm{Screw}_{\hat{x}_{i-1}}(a_i, \alpha_i)\ \mathrm{Screw}_{\hat{z}_i}(d_i, \theta_i) \quad (2)$$

$$ \mathrm{Screw}_{\hat{z}_i}(d_i, \theta_i) = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & 0 \\ s\theta_i & c\theta_i & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3)$$





$$\text{Screw}_{\hat{x}_{i-1}}(a_i, \alpha_i) = \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & c\alpha_i & -s\alpha_i & 0 \\ 0 & s\alpha_i & c\alpha_i & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4}$$

Since

$$ {}^{i-1}\boldsymbol{T}_i = \mathrm{Screw}_{\hat{x}_{i-1}}(a_i, \alpha_i)\ \mathrm{Screw}_{\hat{z}_i}(d_i, \theta_i) \tag{5} $$

they lead to

$$ {}^{i-1}\boldsymbol{T}_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_i \\ s\theta_i c\alpha_i & c\theta_i c\alpha_i & -s\alpha_i & -s\alpha_i d_i \\ s\theta_i s\alpha_i & c\theta_i s\alpha_i & c\alpha_i & c\alpha_i d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6} $$

Very important to notice that this is a generic matrix assuming translation and rotation happened in the positive direction. You have to consider the sign of parameters in specific cases at the point of substituting values.

The homogeneous transformation matrix is in the form given by

$$ {}^{i-1}\boldsymbol{T}_i = \begin{bmatrix} {}^{i-1}\boldsymbol{R}_i & {}^{i-1}\boldsymbol{p}_i \\ \boldsymbol{O}_{1\times 3} & 1 \end{bmatrix} \tag{7} $$

where

$$ {}^{i-1}\boldsymbol{R}_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 \\ s\theta_i c\alpha_i & c\theta_i c\alpha_i & -s\alpha_i \\ s\theta_i s\alpha_i & c\theta_i s\alpha_i & c\alpha_i \end{bmatrix} \tag{8} $$

and

$$ {}^{i-1}\boldsymbol{p}_i = \begin{bmatrix} a_i \\ -s\alpha_i d_i \\ c\alpha_i d_i \\ 1 \end{bmatrix} \tag{9} $$

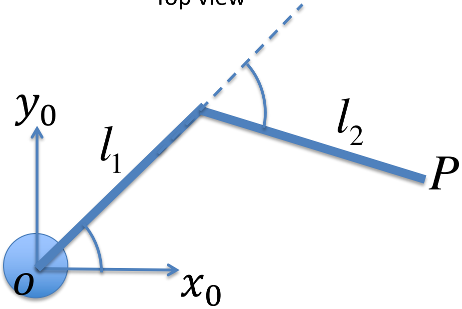
in homogeneous coordinates. Run DHDombreKhalil.m

Example of a Scara robot manipulator

Fill the D-H table for the Scara robot manipulator shown in 10.

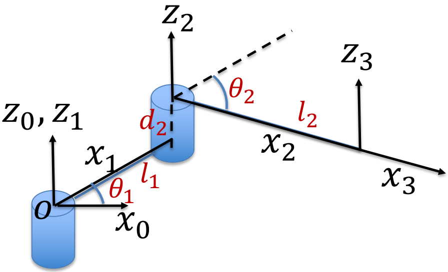
Industrial Scara robot

Top view



We can draw the axes following the above guidelines and fill the DH table given in the figure

Industrial Scara robot



1. a_i = from $\hat{z}_{i-1}$ to $\hat{z}_i$ along $\hat{x}_{i-1}$.
2. α_i = angle $\hat{z}_{i-1}$ and $\hat{z}_i$ about $\hat{x}_{i-1}$.
3. d_i = from $\hat{x}_{i-1}$ to $\hat{x}_i$ along $\hat{z}_i$.
4. θ_i = angle $\hat{x}_{i-1}$ to $\hat{x}_i$ about $\hat{z}_i$.

i	a_i	α_i	d_i	θ_i
1	0	0	0	θ_1
2	l_1	0	d_2	θ_2
3	l_2	0	0	0

1. Find the position of the tip of the manipulator relative to the base frame.
2. What is the orientation of the second link relative to the base frame?

i	a_i	α_i	d_i	θ_i
1	0	0	0	θ_1
2	l_1	0	d_2	θ_2
3	l_2	0	0	0

$$
{}^{i-1}\boldsymbol{T}_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_i \\ s\theta_i c\alpha_i & c\theta_i c\alpha_i & -s\alpha_i & -s\alpha_i d_i \\ s\theta_i s\alpha_i & c\theta_i s\alpha_i & c\alpha_i & c\alpha_i d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{10}
$$

Using the DH table and equation (10) we can write:

$$
{}^{0}\boldsymbol{T}_1 = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{11}
$$

$$
{}^{1}\boldsymbol{T}_2 = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & l_1 \\ s\theta_2 & c\theta_2 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (12)
\qquad
{}^{2}\boldsymbol{T}_3 = \begin{bmatrix} 1 & 0 & 0 & l_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (13)
$$

Run Scara.m to obtain the homogeneous transformation matrix
${}^{0}\boldsymbol{T}_3 =^{0} \boldsymbol{T}_1 \, {}^{1}\boldsymbol{T}_2 \, {}^{2}\boldsymbol{T}_3$

$$
{}^{0}\boldsymbol{T}_3 = \begin{bmatrix} c(\theta_1+\theta_2) & -s(\theta_1+\theta_2) & 0 & l_1 c\theta_1 + l_2 c(\theta_1+\theta_2) \\ s(\theta_1+\theta_2) & c(\theta_1+\theta_2) & 0 & l_1 s\theta_1 + l_2 s(\theta_1+\theta_2) \\ 0 & 0 & 1 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (14)
$$

Since,

$$ {}^{j}\boldsymbol{T}_i = \begin{bmatrix} {}^{j}\boldsymbol{R}_i & {}^{j}\boldsymbol{p}_i \\ \boldsymbol{O}_{1\times 3} & 1 \end{bmatrix} \tag{15} $$

the position vector of the end-effector in the base frame (x_0, y_0, z_0) is given by the last column of ${}^{0}\boldsymbol{T}_3$:

$$ P = \begin{bmatrix} l_1 c\theta_1 + l_2 c(\theta_1 + \theta_2) \\ l_1 s\theta_1 + l_2 s(\theta_1 + \theta_2) \\ d_2 \\ 1 \end{bmatrix} \tag{16} $$

Very important: Inspection of Scara robot shows that θ_1 is positive and θ_2 is negative in this particular configuration. However, DH-parameters give a generic equation of the position vector of the end-effector in the base frame (x_0, y_0, z_0). This means that you have to substitute the exact values of θ_1 and θ_2 with their signs to obtain the numerical values of the elements in the position vector.

To obtain the orientation between different axes, we use our knowledge about the orientation matrix obtained using projections between unit vectors:

$$^{0}\boldsymbol{R}_3 = \begin{bmatrix} \hat{\boldsymbol{x}}_3 \cdot \hat{\boldsymbol{x}}_0 & \hat{\boldsymbol{y}}_3 \cdot \hat{\boldsymbol{x}}_0 & \hat{\boldsymbol{z}}_3 \cdot \hat{\boldsymbol{x}}_0 \\ \hat{\boldsymbol{x}}_3 \cdot \hat{\boldsymbol{y}}_0 & \hat{\boldsymbol{y}}_3 \cdot \hat{\boldsymbol{y}}_0 & \hat{\boldsymbol{z}}_3 \cdot \hat{\boldsymbol{y}}_0 \\ \hat{\boldsymbol{x}}_3 \cdot \hat{\boldsymbol{z}}_0 & \hat{\boldsymbol{y}}_3 \cdot \hat{\boldsymbol{z}}_0 & \hat{\boldsymbol{z}}_3 \cdot \hat{\boldsymbol{z}}_0 \end{bmatrix} \tag{17}$$

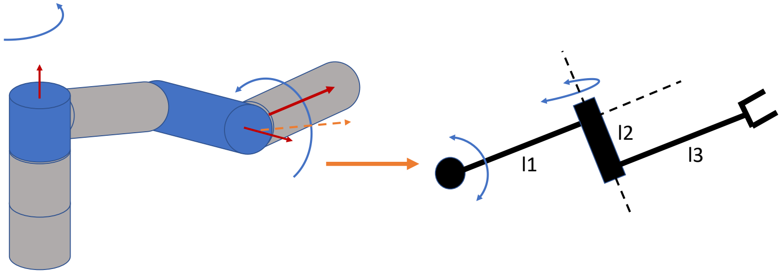
$$^{0}\boldsymbol{R}_3 = \begin{bmatrix} c(\theta_1 + \theta_2) & -s(\theta_1 + \theta_2) & 0 \\ s(\theta_1 + \theta_2) & c(\theta_1 + \theta_2) & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{18}$$

The orientation of the last link relative to the base frame is given by $\arccos(\hat{x}_3 \cdot \hat{x}_0) = (\theta_1 + \theta_2)$

This orientation is given as a generic equation. You have to substitute the exact values of θ_1 and θ_2 with their signs to obtain the numerical value of orientation. Similarly check if the angles between other pairs of axes make sense.

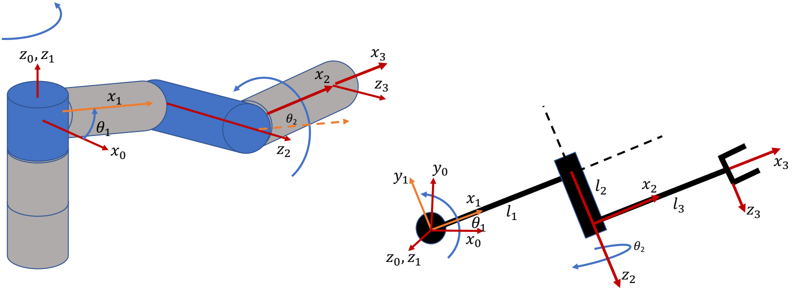
Example for manipulators with perpendicular axes

Assign the frames following the D-H convention of Khalil and Dombre for the following manipulator. Hence find the position of the end point relative to the top of the first vertical link.



1. a_i = the distance from $\hat{z}_{i-1}$ to $\hat{z}_i$ measured along $\hat{x}_{i-1}$.
2. α_i = the angle between $\hat{z}_{i-1}$ and $\hat{z}_i$ measured about $\hat{x}_{i-1}$.
3. d_i = the distance from $\hat{x}_{i-1}$ to $\hat{x}_i$ measured along $\hat{z}_i$.
4. θ_i = the angle between $\hat{x}_{i-1}$ to $\hat{x}_i$ measured about $\hat{z}_i$.

Alternative 1



Then the D-H table is given by

1. a_i = from $\hat{z}_{i-1}$ to $\hat{z}_i$ along $\hat{x}_{i-1}$.
2. α_i = angle $\hat{z}_{i-1}$ and $\hat{z}_i$ about $\hat{x}_{i-1}$.
3. d_i = from $\hat{x}_{i-1}$ to $\hat{x}_i$ along $\hat{z}_i$.
4. θ_i = angle $\hat{x}_{i-1}$ to $\hat{x}_i$ about $\hat{z}_i$.

i	a_i	α_i	d_i	θ_i
1	0	0	0	θ_1
2	l_1	90	l_2	θ_2
3	l_3	0	0	0

i	a_i	α_i	d_i	θ_i
1	0	0	0	θ_1
2	l_1	90	l_2	θ_2
3	l_3	0	0	0

$$
{}^{i-1}\boldsymbol{T}_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_i \\ s\theta_i c\alpha_i & c\theta_i c\alpha_i & -s\alpha_i & -s\alpha_i d_i \\ s\theta_i s\alpha_i & c\theta_i s\alpha_i & c\alpha_i & c\alpha_i d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{19}
$$

Then we can use the DH table and equation (24) to write down the transformation matrices given by,

$$
{}^{0}\boldsymbol{T}_1 = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{20}
$$

$$
{}^{1}\boldsymbol{T}_2 = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & l_1 \\ 0 & 0 & -1 & -l_2 \\ s\theta_2 & c\theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{21}
$$

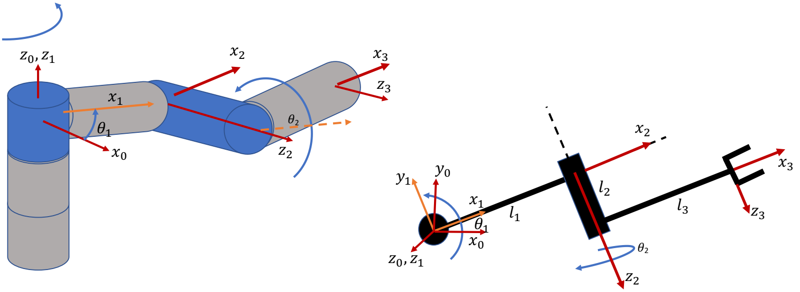
$$
{}^{2}\boldsymbol{T}_3 = \begin{bmatrix} 1 & 0 & 0 & l_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{22}
$$

${}^{0}\boldsymbol{T}_3 =^{0} \boldsymbol{T}_1 \; {}^{1}\boldsymbol{T}_2 \; {}^{2}\boldsymbol{T}_3$

$$
{}^{0}\boldsymbol{T}_3 = \begin{bmatrix} c\theta_1 c\theta_2 & -c\theta_1 s\theta_2 & s\theta_1 & l_1 c\theta_1 + l_2 s\theta_1 + l_3 c\theta_1 c\theta_2 \\ s\theta_1 c\theta_2 & -s\theta_1 s\theta_2 & -c\theta_1 & l_1 s\theta_1 - l_2 c\theta_1 + l_3 c\theta_2 s\theta_1 \\ s\theta_2 & c\theta_2 & 0 & l_3 s\theta_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{23}
$$

Run ExamplewithPerpAxes.m Alternative-1.

Alternative 2



Then the D-H table is given by

1. a_i = from $\hat{z}_{i-1}$ to $\hat{z}_i$ along $\hat{x}_{i-1}$.
2. α_i = angle $\hat{z}_{i-1}$ and $\hat{z}_i$ about $\hat{x}_{i-1}$.
3. d_i = from $\hat{x}_{i-1}$ to $\hat{x}_i$ along $\hat{z}_i$.
4. θ_i = angle $\hat{x}_{i-1}$ to $\hat{x}_i$ about $\hat{z}_i$.

i	a_i	α_i	d_i	θ_i
1	0	0	0	θ_1
2	l_1	90	0	θ_2
3	l_3	0	l_2	0

i	a_i	α_i	d_i	θ_i
1	0	0	0	θ_1
2	l_1	90	0	θ_2
3	l_3	0	l_2	0

$$
{}^{i-1}\boldsymbol{T}_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_i \\ s\theta_i c\alpha_i & c\theta_i c\alpha_i & -s\alpha_i & -s\alpha_i d_i \\ s\theta_i s\alpha_i & c\theta_i s\alpha_i & c\alpha_i & c\alpha_i d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (24)
$$

Then we can use the DH table and equation (24) to write down the transformation matrices given by,

$$
{}^{0}\boldsymbol{T}_1 = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (25)
$$

$$
{}^{1}\boldsymbol{T}_2 = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & l_1 \\ 0 & 0 & -1 & 0 \\ s\theta_2 & c\theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{26}
$$

$$
{}^{2}\boldsymbol{T}_3 = \begin{bmatrix} 1 & 0 & 0 & l_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & l_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{27}
$$

${}^{0}\boldsymbol{T}_3 =^{0} \boldsymbol{T}_1 \, {}^{1}\boldsymbol{T}_2 \, {}^{2}\boldsymbol{T}_3$

$$
{}^{0}\boldsymbol{T}_3 = \begin{bmatrix} c\theta_1 c\theta_2 & -c\theta_1 s\theta_2 & s\theta_1 & l_1 c\theta_1 + l_2 s\theta_1 + l_3 c\theta_1 c\theta_2 \\ s\theta_1 c\theta_2 & -s\theta_1 s\theta_2 & -c\theta_1 & l_1 s\theta_1 - l_2 c\theta_1 + l_3 c\theta_2 s\theta_1 \\ s\theta_2 & c\theta_2 & 0 & l_3 s\theta_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{28}
$$

Run ExamplewithPerpAxes.m Alternative-2.

since

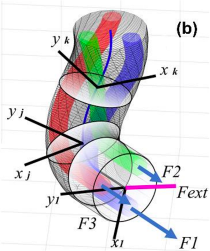
$$ {}^{j}\boldsymbol{T}_i = \begin{bmatrix} {}^{j}\boldsymbol{R}_i & {}^{j}\boldsymbol{p}_i \\ \boldsymbol{O}_{1\times 3} & 1 \end{bmatrix} \tag{29} $$

the position of the end point relative to the top of the first vertical link is given by

$$ {}^{0}\boldsymbol{P}_3 = \begin{bmatrix} l_1 c\theta_1 + l_2 s\theta_1 + l_3 c\theta_1 c\theta_2 \\ l_1 s\theta_1 - l_2 c\theta_1 + l_3 c\theta_2 s\theta_1 \\ l_3 s\theta_2 \end{bmatrix} \tag{30} $$

Again, please note that DH-parameters give a generic equation of the position vector of the end-effector in the base frame (x_0, y_0, z_0). You have to substitute the exact values of θ_1 and θ_2 with their signs to obtain the numerical values of the elements in the position vector.

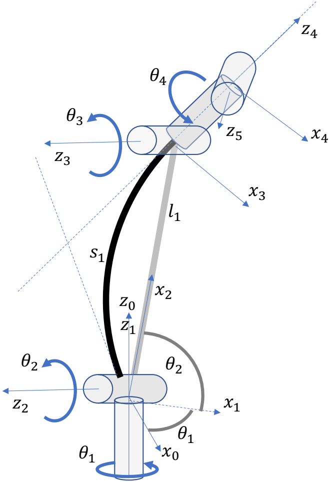
Continuum robots



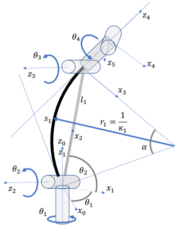
Sadati, S. H., Shiva, A., Herzig, N., Rucker, C. D., Hauser, H., Walker, I. D., ... & Nanayakkara, T. (2020). Stiffness Imaging With a Continuum Appendage: Real-Time Shape and Tip Force Estimation From Base Load Readings. *IEEE Robotics and Automation Letters*, *5*(2), 2824-2831.

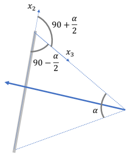
Continuum robots are inspired by biological inspirations such as the elephant trunk, octopus arms, and chameleon tongue. They are still under research for applications such as minimally invasive surgery, malleable robotic arms, and fingers for soft object manipulation.

Continuum robots - for your knowledge only



Continuum robots - for your knowledge only





1. a_i = from $\hat{z}_{i-1}$ to $\hat{z}_i$ along $\hat{x}_{i-1}$.
2. α_i = angle $\hat{z}_{i-1}$ and $\hat{z}_i$ about $\hat{x}_{i-1}$.
3. d_i = from $\hat{x}_{i-1}$ to $\hat{x}_i$ along $\hat{z}_i$.
4. θ_i = angle $\hat{x}_{i-1}$ to $\hat{x}_i$ about $\hat{z}_i$.

Then the D-H table is given by

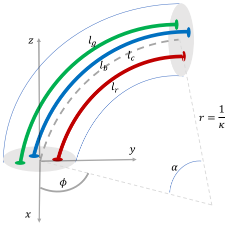
i	a_i	α_i	d_i	θ_i
1	0	0	0	θ_1
2	0	90	0	θ_2
3	l_1	0	0	$-\left(90 + \frac{\alpha}{2}\right)$
4	0	-90	0	θ_4

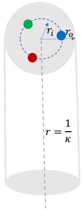
$$\alpha = \frac{s_1}{r_1} = s_1 \kappa_1$$

$$l_1 = 2 r_1 \sin\frac{\alpha}{2} = \frac{2}{\kappa_1} \sin\frac{s_1 \kappa_1}{2}$$

Run Continuumseg.m

Tendon driven continuum robots - for your knowledge only





$$l_c = \frac{l_r + l_b + l_g}{3}$$

$$\phi = \tan^{-1}\left(\frac{l_g + l_b - 2l_r}{\sqrt{3}\,(l_g - l_b)}\right)$$

$$\kappa = 2\frac{\sqrt{l_r^2 + l_g^2 + l_b^2 - l_r l_g - l_g l_b - l_r l_b}}{r_i\,(l_r + l_g + l_b)}$$

$$\alpha = \frac{l_c}{r} = l_c \kappa$$

For derivation, please see:

Jones, B. A., Walker, I. D. (2006). Kinematics for multisection continuum robots. IEEE Transactions on Robotics, 22(1), 43-55.